

Exercício $A = \{0^n 10^n \mid n \geq 0\}$

$p=4$

$S = 0^n 10^n \mid S = 0^4 \mid 0^4 \mid = 0000 \mid 0000 \mid$

case #1

$\underbrace{00}_{x} \underbrace{0010000}_{y} \underbrace{1}_{z} xy^2z$

00 00 1 00 00 00 00

$4=4$

$|y| > p \rightarrow |x| > 4$

$|xy| \leq p \rightarrow 7 \leq 4$

case #2

$\underbrace{000}_{x} \underbrace{0100001}_{y} xy^2z$

000 010 010 0001

$4 \neq 7$

$|y| > p \rightarrow |z| > 4$

$|xy| \leq p \rightarrow 6 \leq 4$

case #3

$\underbrace{00}_{x} \underbrace{0010000}_{y} \underbrace{1}_{z} xy^2z$

00 00 00 100001

$6 \neq 4$

$|y| > p \rightarrow |z| > 4$

$|xy| \leq p \rightarrow 4 \leq 4$